Muhammad Hassan Umar

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Mechanical engineering student at the University of Wollongong Dubai (92 WAM) with industry experience from a Mercedes-Benz internship in data analysis and Power BI dashboard development. Seeking research and internship roles in mechanical and aerospace engineering.

EDUCATION

University of Wollongong Dubai — Dubai | UAE
Bachelor of Engineering (Honours) in Mechanical Engineering — *Apr 2025 – Mar 2029*
- Dean's List Recipient (2025) – Top 5% of students.
- 92 WAM (High Distinction Average).
- Relevant Coursework: Fundamentals of Engineering Mechanics, Materials in Design, Engineering Design for Sustainability, Electrical Systems, Advanced Engineering Mathematics, Engineering Fluid Mechanics.

Karlsruhe Institute of Technology — Karlsruhe | Germany
Bachelor of Science in Mechanical Engineering — *Oct 2023 – Mar 2025*
- Relevant Coursework: Mechanical Design I & II, Materials Science, Global Production and Management, Basics of Manufacturing Technology.

PROFESSIONAL EXPERIENCE

Mercedes-Benz AG — Sindelfingen | Germany
Engineering Intern — *Nov 2024 – Jan 2025*
- Designed and developed a 4-tier Power BI dashboard system enabling real-time scrap visibility, projecting a 63% reduction in scrap costs in year 1 and 74-83% over 4-5 years.
- Applied moving averages and Pareto weighting to production defect data to identify and prioritize highest-impact scrap sources across production lines.
- Presented findings to board-level stakeholders in a thesis-style defense as part of a team of 3; placed first among competing project groups and received a 1.3.

PROJECTS

Stellum: Sustainable Manufacturing Plant Design — Dubai | UAE
Engineering Design for Sustainability — *Grade: 98*
- Designed a sustainable production facility in Vietnam within a $10M CAPEX. Delivered at $9.5M with a 2.2-year payback period, $18.43M NPV and 45.46% ROI.
- Produced AutoCAD drawings of a six-zone factory layout and stainless-steel product design selected for minimal material complexity and full end-of-life recyclability.

IB Extended Essay: Aerodynamic Wing Lift Analysis — Sohar | Oman
Independent Research
- Fabricated a wind tunnel with contraction and diffuser cones to investigate the effect of wingspan on lift forces across 5 rectangular wing profiles at optimum angle of attack.
- Collected and processed 30,000 datapoints using a Vernier force sensor and LoggerPro; computed lift coefficients, induced drag, and lift-to-drag ratios to verify aspect ratio.

Projectile Motion Trajectory Analyzer — Dubai | UAE
Engineering Computing and Analysis — *Grade: 100*
- Developed a MATLAB App Designer application computing minimum launch angles and velocities for basketball trajectories from user defined parameters.
- Generated interactive 3D trajectory plots with real-time collision detection, visualizing valid and invalid throw paths against user-specified building parameters.

SKILLS AND PERSONAL

Languages: English (Native), Urdu (Fluent), French (Proficient), Arabic (Conversational), German (Basic)
Technical Skills: AutoCAD | MATLAB | Power BI | Microsoft 365 | Azure | JavaScript | HTML | CSS